

[Embedded Systems-Lab]

[Design Project Report]



[INTEGRATED SMART WORKSPACE SAFETY STATION]

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INTRODUCTION

1.1 Background of the Project

In modern industrial and laboratory environments, safety is a very important concern. Many accidents happen not because of complex failures, but due to simple environmental issues such as low lighting or high temperature. Poor lighting can reduce visibility and increase the chances of human error, while high temperature can damage equipment or even cause fire hazards. Therefore, continuous monitoring of these environmental conditions is necessary.

Traditionally, these conditions are checked manually using separate instruments. This approach is not efficient because it requires constant human attention and does not provide instant alerts. In critical situations, even a small delay can lead to serious consequences. With the advancement of embedded systems, it is now possible to design smart devices that can monitor conditions in real time and respond immediately.

This project focuses on developing a compact and standalone system using the TM4C123GH6PM microcontroller and BOOSTXL-EDUMKII BoosterPack. The system uses built-in sensors to measure light intensity and temperature continuously. Based on these readings, it provides feedback using LEDs, a buzzer, and a serial display. The aim is to create a reliable safety station that works automatically without needing constant human supervision.

1.2 Purpose of the System

The main purpose of this project is to design a smart workspace safety station that can detect unsafe environmental conditions and alert the user instantly. The system is designed to be simple, fast, and effective in providing real-time feedback.

The system monitors two main parameters: light intensity and temperature. These values are compared with predefined safe limits. Depending on the condition, the system classifies the environment into three zones: Safe, Caution, and Danger. Each zone has a specific response using LED colors and buzzer sounds. For example, green light indicates safe conditions, while red light with buzzer indicates danger.

Another important purpose of this system is user interaction. The system includes buttons to reset alarms and switch between active monitoring and maintenance mode. This makes the system flexible and user-friendly. Additionally, the system sends live data to a computer using UART, allowing remote monitoring through software like Tera Term.

Overall, the goal is to build a low-cost, real-time safety monitoring system that improves awareness and reduces risks in working environments.

PROBLEM ANALYSIS

2.1 Identification of the Problem

In many workplaces such as labs, workshops, and industrial setups, environmental conditions are often ignored until a problem occurs. Workers usually rely on manual observation to check whether the lighting is sufficient or if the temperature is within a safe range. This method is not reliable because humans cannot continuously monitor these parameters with accuracy.

One major issue is poor lighting. When the light level is too low, workers may not clearly see their tasks, which can lead to mistakes or accidents. Similarly, high temperature is another serious problem. Equipment can overheat due to continuous usage, poor ventilation, or environmental conditions. Overheating can damage components, reduce system performance, or even cause dangerous situations like fire.

Another problem is the lack of immediate alerts. Even if a worker notices a problem, there may be a delay in taking action. In critical systems, this delay can lead to significant damage. Moreover, there is no centralized system that provides continuous monitoring along with instant feedback.

2.2 Limitations of Existing Methods

Existing methods for monitoring environmental conditions have several limitations. First, manual checking is time-consuming and not reliable. A person cannot monitor light and temperature levels all the time, especially in large or busy environments. This increases the risk of missing critical changes.

Second, separate devices are often used for measuring temperature and light. These devices do not communicate with each other and do not provide a combined safety status. This makes it difficult to understand the overall condition of the workspace.

Another limitation is the lack of real-time alerts. Many traditional systems only display readings but do not provide immediate warnings when conditions become unsafe. Without proper alerts, users may not respond quickly enough to prevent accidents.

Additionally, most systems do not include user interaction features such as alarm reset or mode switching. This reduces flexibility and control for the user.

Finally, remote monitoring is often not available in basic systems. Supervisors cannot check the system status from a distance, which limits the usefulness of the system in larger setups.

Because of these limitations, there is a need for an integrated system that combines sensing, processing, alerting, and user interaction in a single platform.

DESIGN REQUIREMENTS

3.1 Functional Requirements

The system must perform several important functions to meet the project objectives. First, it should continuously monitor environmental conditions, specifically light intensity and temperature. This requires proper integration of the built-in OPT3001 light sensor and TMP006 temperature sensor available on the BoosterPack. The system must read sensor values in real time without interruption.

Second, the system must classify the environment into three zones: Safe, Caution, and Danger. This classification should be based on predefined threshold values for both light and temperature. For example, sufficient light and normal temperature should result in a Safe zone, while extreme values should trigger a Danger zone.

Third, the system must provide immediate feedback to the user. This includes visual feedback using RGB LEDs and audio feedback using a buzzer. Each zone should have a distinct indication, such as green LED for safe, blue LED for caution, and red LED for danger. The buzzer should produce different beep patterns depending on the severity level.

Another key requirement is user interaction. The system must allow the user to reset alarms using a button and switch between Active Monitoring mode and Maintenance mode using another button. Maintenance mode should pause all monitoring and alerts.

3.2 Non-Functional Requirements

In addition to functional requirements, the system must also meet certain non-functional requirements. One important requirement is reliability. The system should work continuously without crashing or missing sensor readings. Since it is a safety system, consistent performance is very important.

Another requirement is real-time response. The system should quickly react to changes in light and temperature. There should be minimal delay between sensing the data and generating alerts. This ensures that the user is informed immediately about unsafe conditions.

The system should also be simple and user-friendly. The LED colors and buzzer sounds should be easy to understand, so that even a non-technical user can quickly recognize the system status. The button functions should also be straightforward and responsive.

Power efficiency is another consideration. The system should not consume excessive power, especially since it runs continuously. Using efficient code and proper control of peripherals helps in reducing power usage.

POSSIBLE SOLUTIONS

4.1 Hardware-Based Solutions

One possible solution to the problem is to design a fully hardware-based system using sensors, microcontrollers, and output devices. In this approach, sensors like light-dependent resistors (LDRs) and analog temperature sensors could be used instead of digital sensors. These sensors would provide analog signals that can be processed by a microcontroller using ADC (Analog-to-Digital Converter).

The microcontroller, such as the TM4C123GH6PM, can then process these signals and determine the safety status. Based on the result, it can control LEDs and a buzzer to alert the user. This type of system is simple and cost-effective, but it may not provide very accurate readings compared to digital sensors.

Another hardware-based option is to use external modules for display, such as LCD screens, to show real-time values directly on the device. However, this increases system complexity and cost. Also, additional wiring and interfacing are required.

While hardware-only solutions are useful, they may lack flexibility and advanced features like remote monitoring and easy updates.

4.2 Embedded System-Based Solution (Chosen Approach)

The selected solution for this project is an embedded system-based design using the Tiva C LaunchPad and BOOSTXL-EDUMKII BoosterPack. This approach combines both hardware and software to create a more efficient and flexible system.

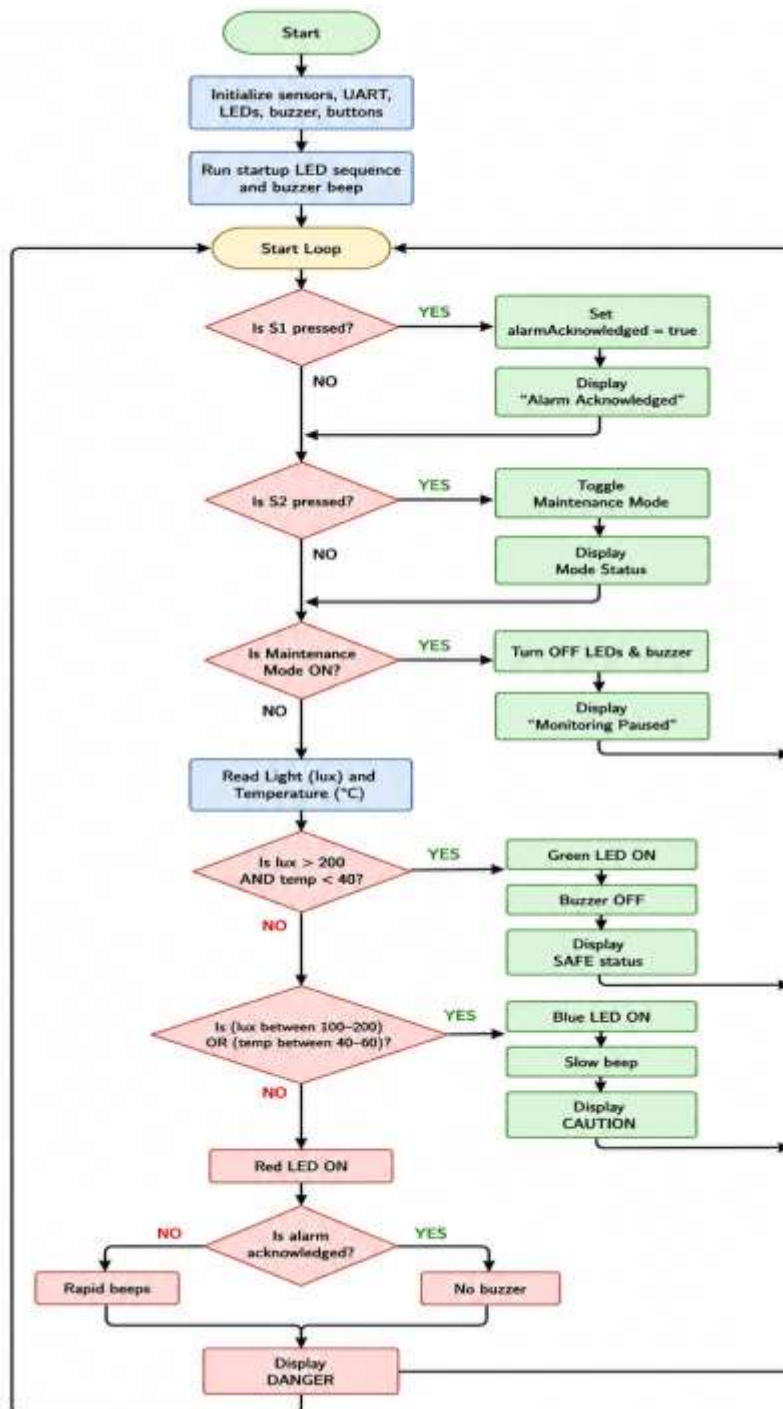
The BoosterPack already includes high-accuracy digital sensors (OPT3001 and TMP006), which communicate with the microcontroller. In your implementation, these sensors are accessed through libraries in Energia, making the design simpler and faster to develop.

The microcontroller processes the sensor data in real time and compares it with predefined thresholds. Based on the result, it controls the RGB LEDs and buzzer to provide clear alerts. The use of software allows easy modification of threshold values and behavior without changing hardware.

Another advantage of this approach is UART communication. The system sends live data to a PC, allowing remote monitoring using Tera Term. This feature is not easily achievable in basic hardware-only systems.

Overall, this embedded system-based solution provides better accuracy, flexibility, and scalability. It meets all the project requirements while keeping the design efficient and easy to implement.

PRELIMINARY DESIGN



DESIGN DESCRIPTION

6.1 Hardware Design Description

The hardware design of this system is based on the TM4C123GH6PM microcontroller mounted on the Tiva C LaunchPad. The BOOSTXL-EDUMKII BoosterPack is connected directly to the LaunchPad, which provides built-in sensors and simplifies interfacing.

The system uses two main sensors: OPT3001 for light measurement and TMP006 for temperature measurement. These sensors are already integrated into the BoosterPack, so no external wiring is required. This reduces complexity and improves reliability.

For output, the system uses an RGB LED and a piezo buzzer. The RGB LED is connected to specific GPIO pins (PF_1, PF_3, and PE_4), and it provides visual indication using different colors. The buzzer is connected to another GPIO pin (PF_2) and generates sound alerts.

Two push buttons are used for user interaction. These buttons are connected to PF_4 and PF_0 and are configured using internal pull-up resistors. This means the button is active when pressed (logic LOW).

All components are powered through the LaunchPad, making the system compact and easy to use. The hardware setup is simple because most components are already available on the BoosterPack.

6.2 Software Design Description

The software design is implemented using Energia IDE, which simplifies programming for the Tiva C microcontroller. The code is structured into two main functions: `setup()` and `loop()`.

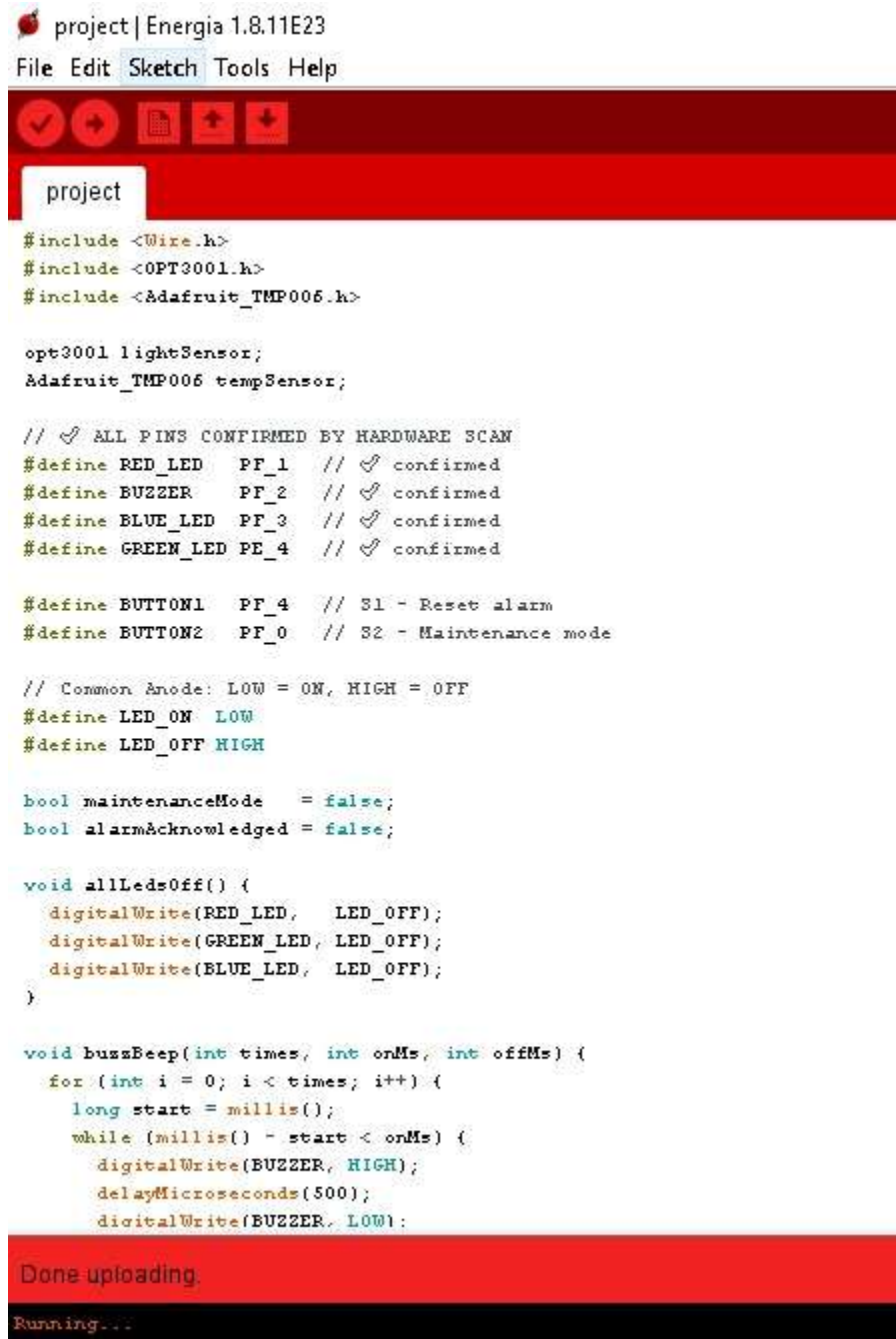
In the `setup()` function, all peripherals are initialized. This includes starting serial communication, initializing sensors, and configuring GPIO pins for LEDs, buzzer, and buttons. A startup sequence is also included to verify that all components are working correctly.

The `loop()` function contains the main logic of the system. It continuously checks button inputs and updates system states such as maintenance mode and alarm acknowledgment. After handling user input, the system reads sensor data for light and temperature.

The sensor values are then compared with predefined thresholds to determine the safety zone. Based on the result, the system controls LEDs and buzzer accordingly. For example, in the danger zone, the buzzer produces rapid beeps unless the alarm is acknowledged.

The software also sends real-time data to the serial monitor (Tera Term), allowing remote observation of system status. Delays are used carefully to maintain stability while ensuring the system remains responsive.

SOFTWARE SIMULATION



The screenshot displays the Energia IDE interface. At the top, the title bar reads 'project | Energia 1.8.11E23'. Below it is a menu bar with 'File', 'Edit', 'Sketch', 'Tools', and 'Help'. A toolbar contains icons for opening, saving, and running. The main workspace shows a C++ sketch for a hardware simulation. The sketch includes headers for `Wire.h`, `OPT3001.h`, and `Adafruit_TMP006.h`. It defines pins for LEDs, a buzzer, and buttons, with comments indicating they are confirmed by hardware scan. It also defines constants for LED states and button functions. The sketch includes functions for turning all LEDs off and for a buzzer beep sequence.

```
project | Energia 1.8.11E23
File Edit Sketch Tools Help

#include <Wire.h>
#include <OPT3001.h>
#include <Adafruit_TMP006.h>

opt3001 lightSensor;
Adafruit_TMP006 tempSensor;

// ♥ ALL PINS CONFIRMED BY HARDWARE SCAN
#define RED_LED PF_1 // ♥ confirmed
#define BUZZER PF_2 // ♥ confirmed
#define BLUE_LED PF_3 // ♥ confirmed
#define GREEN_LED PE_4 // ♥ confirmed

#define BUTTON1 PF_4 // S1 - Reset alarm
#define BUTTON2 PF_0 // S2 - Maintenance mode

// Common Anode: LOW = ON, HIGH = OFF
#define LED_ON LOW
#define LED_OFF HIGH

bool maintenanceMode = false;
bool alarmAcknowledged = false;

void allLedsOff() {
    digitalWrite(RED_LED, LED_OFF);
    digitalWrite(GREEN_LED, LED_OFF);
    digitalWrite(BLUE_LED, LED_OFF);
}

void buzzBeep(int times, int onMs, int offMs) {
    for (int i = 0; i < times; i++) {
        long start = millis();
        while (millis() - start < onMs) {
            digitalWrite(BUZZER, HIGH);
            delayMicroseconds(500);
            digitalWrite(BUZZER, LOW);
        }
    }
}
```

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Running...

project | Energia 1.8.11E23

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project

```
        delayMicroseconds(500);
    }
    if (i < times - 1) delay(offMs);
}

void setup() {
    Serial.begin(9600);
    Wire.begin();
    lightSensor.begin();
    tempSensor.begin();

    pinMode(RED_LED,    OUTPUT);
    pinMode(GREEN_LED,  OUTPUT);
    pinMode(BLUE_LED,   OUTPUT);
    pinMode(BUZZER,     OUTPUT);
    pinMode(BUTTON1,    INPUT_PULLUP);
    pinMode(BUTTON2,    INPUT_PULLUP);

    allLedsOff();
    digitalWrite(BUZZER, LOW);

    // Startup animation - confirms all hardware working
    Serial.println("=== Smart Workspace Safety Station ===");
    Serial.println("Running startup check...");

    digitalWrite(RED_LED,  LED_ON); delay(500);
    digitalWrite(RED_LED,  LED_OFF);
    digitalWrite(GREEN_LED, LED_ON); delay(500);
    digitalWrite(GREEN_LED, LED_OFF);
    digitalWrite(BLUE_LED, LED_ON); delay(500);
    digitalWrite(BLUE_LED, LED_OFF);
    buzzBeep(1, 100, 0);

    Serial.println("Startup OK. Monitoring started.");
}
```

Done uploading.

Running...

project | Energia 1.8.11E23

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```
void loop() {  
  // S1 - Acknowledge/reset alarm  
  if (digitalRead(BUTTON1) == LOW) {  
    delay(50);  
    alarmAcknowledged = true;  
    Serial.println("[BTN1] Alarm acknowledged.");  
    while (digitalRead(BUTTON1) == LOW);  
  }  
  
  // S2 - Toggle maintenance mode  
  if (digitalRead(BUTTON2) == LOW) {  
    delay(50);  
    maintenanceMode = !maintenanceMode;  
    alarmAcknowledged = false;  
    Serial.println(maintenanceMode ? "[MODE] MAINTENANCE" : "[MODE] ACTIVE MONITORING");  
    while (digitalRead(BUTTON2) == LOW);  
  }  
  
  // Maintenance mode - all off, no monitoring  
  if (maintenanceMode) {  
    allLedsOff();  
    digitalWrite(BUZZER, LOW);  
    Serial.println("[MAINTENANCE] Monitoring paused.");  
    delay(1000);  
    return;  
  }  
  
  // Read sensors  
  float lux = lightSensor.readResult();  
  float temp = tempSensor.readObjTempC();  
  
  // Zone detection and response  
  if (lux > 200 && temp < 40) {  
    // SAFE ZONE - Green LED, silent  
    allLedsOff();  
  }  
}
```

Done uploading.

Running...

project | Energia 1.8.11E23

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project

```
digitalWrite(BUZZER, LOW);
Serial.println("[MAINTENANCE] Monitoring paused.");
delay(1000);
return;
}

// Read sensors
float lux = lightSensor.readResult();
float temp = tempSensor.readObjTempC();

// Zone detection and response
if (lux > 200 && temp < 40) {
    // SAFE ZONE - Green LED, silent
    allLedsOff();
    digitalWrite(GREEN_LED, LED_ON);
    digitalWrite(BUZZER, LOW);
    alarmAcknowledged = false;
    Serial.print("[SAFE] ");
} else if ((lux >= 100 && lux <= 200) || (temp >= 40 && temp < 50)) {
    // CAUTION ZONE - Blue LED, single slow beep
    allLedsOff();
    digitalWrite(BLUE_LED, LED_ON);
    buzzBeep(1, 200, 0);
    Serial.print("[CAUTION] ");
} else {
    // DANGER ZONE - Red LED, rapid triple beep
    allLedsOff();
    digitalWrite(RED_LED, LED_ON);
    if (!alarmAcknowledged) {
        buzzBeep(3, 100, 80);
    }
    Serial.print("[DANGER] ");
}
```

Done uploading.

Running...

🍏 project | Energia 1.8.11E23

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project

```
// Read sensors
float lux = lightSensor.readResult();
float temp = tempSensor.readObjTempC();

// Zone detection and response
if (lux > 200 && temp < 40) {
  // SAFE ZONE - Green LED, silent
  allLedsOff();
  digitalWrite(GREEN_LED, LED_ON);
  digitalWrite(BUZZER, LOW);
  alarmAcknowledged = false;
  Serial.print("[SAFE] ");

} else if ((lux >= 100 && lux <= 200) || (temp >= 40 && temp < 60)) {
  // CAUTION ZONE - Blue LED, single slow beep
  allLedsOff();
  digitalWrite(BLUE_LED, LED_ON);
  buzzBeep(1, 200, 0);
  Serial.print("[CAUTION] ");

} else {
  // DANGER ZONE - Red LED, rapid triple beep
  allLedsOff();
  digitalWrite(RED_LED, LED_ON);
  if (!alarmAcknowledged) {
    buzzBeep(3, 100, 80);
  }
  Serial.print("[DANGER] ");
}

Serial.print("Light: "); Serial.print(lux);
Serial.print(" lx | Temp: "); Serial.print(temp);
Serial.println(" C");

delay(500);
}
```

Done uploading.

Running...

safe (no contact)

COM4			
[SAFE]	Light:	4076.00 lx	Temp: 27.95 C
[SAFE]	Light:	4076.00 lx	Temp: 29.84 C
[SAFE]	Light:	4072.00 lx	Temp: 29.84 C
[SAFE]	Light:	4076.00 lx	Temp: 29.84 C
[SAFE]	Light:	4076.00 lx	Temp: 29.84 C
[SAFE]	Light:	4076.00 lx	Temp: 29.84 C
[SAFE]	Light:	4072.00 lx	Temp: 29.84 C
[SAFE]	Light:	4076.00 lx	Temp: 29.84 C
[SAFE]	Light:	4076.00 lx	Temp: 28.36 C
[SAFE]	Light:	4073.00 lx	Temp: 28.36 C
[SAFE]	Light:	4076.00 lx	Temp: 28.36 C
[SAFE]	Light:	3988.00 lx	Temp: 28.36 C
[SAFE]	Light:	3988.00 lx	Temp: 28.36 C
[SAFE]	Light:	4068.00 lx	Temp: 28.36 C
[SAFE]	Light:	4076.00 lx	Temp: 28.36 C
[SAFE]	Light:	4068.00 lx	Temp: 28.36 C
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cation and danger (contacted)

COM4			
[SAFE]	Light:	4064.00 lx	Temp: 28.63 C
[SAFE]	Light:	4064.00 lx	Temp: 28.63 C
[SAFE]	Light:	4068.00 lx	Temp: 29.28 C
[CAUTION]	Light:	115.00 lx	Temp: 29.28 C
[SAFE]	Light:	4081.00 lx	Temp: 29.28 C
[DANGER]	Light:	75.00 lx	Temp: 29.28 C
[SAFE]	Light:	4085.00 lx	Temp: 29.28 C
[CAUTION]	Light:	105.00 lx	Temp: 29.28 C
[DANGER]	Light:	96.00 lx	Temp: 29.96 C
[SAFE]	Light:	4036.00 lx	Temp: 29.96 C
[DANGER]	Light:	98.00 lx	Temp: 29.96 C
[SAFE]	Light:	4056.00 lx	Temp: 29.96 C
[SAFE]	Light:	4073.00 lx	Temp: 29.96 C
[SAFE]	Light:	4073.00 lx	Temp: 29.96 C
[SAFE]	Light:	4077.00 lx	Temp: 29.19 C
[SAFE]	Light:	4052.00 lx	Temp: 29.19 C
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EXPERIMENTAL RESULTS

7.1 System Testing and Observations

The system was tested using the Tiva C LaunchPad along with the BOOSTXL-EDUMKII BoosterPack. The program was uploaded using Energia IDE, and the output was monitored through serial communication using Tera Term. Different environmental conditions were created manually to observe how the system responds.

The light and temperature conditions were varied. For testing low light, the sensor was covered partially or fully using a hand. For high temperature, the sensor was exposed to a warm surface or held for a few seconds to increase the reading. These changes were reflected on Tera Term.

The system correctly identified all three zones:

- In normal conditions, the **Safe zone** was activated, and the green LED remained ON with no buzzer sound.
- When light decreased slightly or temperature increased moderately, the **Caution zone** was triggered. The blue LED turned ON, and a slow beep was generated.
- In extreme conditions (very low light or high temperature), the **Danger zone** was activated. The red LED turned ON, and rapid buzzer beeps were observed.

Button functionality was also tested. Pressing S1 successfully stopped the buzzer in danger mode by acknowledging the alarm. Pressing S2 switched the system between active monitoring and maintenance mode.

7.2 Output Results (Tera Term Display)

The system continuously sent real-time data to Tera Term using UART communication. The output included the safety status along with light and temperature readings.

Example outputs observed:

- [SAFE] Light: 250 lx | Temp: 32 C
- [CAUTION] Light: 150 lx | Temp: 45 C
- [DANGER] Light: 80 lx | Temp: 65 C

These outputs clearly showed that the system was correctly processing sensor values and assigning the appropriate safety zone. The readings updated every 500 milliseconds, providing near real-time monitoring.

PERFORMANCE ANALYSIS

8.1 System Efficiency and Accuracy

The performance of the system was evaluated based on its responsiveness, accuracy, and reliability. The system showed efficient performance during testing, as it continuously monitored environmental conditions without interruption.

The use of digital sensors (OPT3001 and TMP006) ensured accurate readings for both light and temperature. The values displayed on Tera Term were stable and consistent, with minimal fluctuations. This indicates that the sensors are reliable for real-time monitoring.

The response time of the system was also very good. The delay between sensor reading and output response was very small (approximately within 500 ms). This means the system can quickly detect unsafe conditions and alert the user without delay.

The buzzer and LED responses were synchronized correctly with the detected zones. Each condition triggered the correct output pattern, showing that the logic implementation is accurate.

The system also handled user inputs efficiently. Button presses were detected correctly, and actions such as alarm reset and mode switching worked without errors.

8.2 Limitations and Improvements

Although the system performed well, there are some limitations. One limitation is the use of fixed threshold values for light and temperature. These values may not be suitable for all environments. For example, different workplaces may require different safety limits.

Another limitation is the lack of data storage. The system only displays real-time values but does not store past data. This makes it difficult to analyze trends over time.

The buzzer sound is also basic, as it uses simple tone generation through GPIO toggling. More advanced sound patterns could improve user awareness.

Additionally, the system relies on manual testing for extreme conditions like high temperature. In real-world applications, more controlled testing methods may be required.

For future improvements, adjustable thresholds can be added using buttons or a user interface. Data logging can be implemented using memory modules or PC software. Wireless communication (such as Bluetooth or Wi-Fi) can also be added for remote monitoring without cables.

Overall, the system performs efficiently for its intended purpose, but there is room for enhancement to make it more flexible and advanced.

CONCLUSION

9.1 Summary of the Project

This project successfully designed and implemented an Integrated Smart Workspace Safety Station using the Tiva C LaunchPad and BOOSTXL-EDUMKII BoosterPack. The system was able to continuously monitor environmental conditions, specifically light intensity and temperature, using built-in digital sensors. Based on these readings, the system classified the environment into Safe, Caution, and Danger zones.

The system provided immediate feedback through multiple outputs. The RGB LED displayed different colors to represent each safety level, while the buzzer generated sound alerts based on the severity of the condition. In addition, real-time data was transmitted to a computer using UART communication and displayed on Tera Term, allowing remote monitoring.

User interaction was also successfully implemented using push buttons. The user could reset alarms and switch between active monitoring and maintenance mode. The system operated in a continuous loop, ensuring real-time response to any environmental changes.

All the required objectives of the project were achieved. The hardware worked correctly, and the software logic implemented in Energia performed as expected. The system was simple, efficient, and reliable for basic safety monitoring applications.

9.2 Future Scope and Improvements

Although the system achieved its main goals, there are several possible improvements that can enhance its functionality. One important improvement is the addition of adjustable threshold values. Instead of fixed limits, users could set their own safety levels based on the environment.

Another improvement is data logging. The system can be extended to store sensor data over time, which can help in analyzing trends and detecting long-term issues. This can be done by connecting the system to a computer database or using external storage.

Wireless communication can also be added to make the system more advanced. Technologies like Bluetooth or Wi-Fi can allow remote monitoring without needing a wired connection to a PC.

The alert system can be improved by adding different sound patterns or even voice alerts for better understanding. A display screen such as an LCD can also be added to show readings directly on the device.

In the future, more sensors can be integrated, such as gas sensors or humidity sensors, to make the system more comprehensive. This would make the safety station suitable for a wider range of industrial and laboratory environments.